

Technical note: searching the ASTRA quantile data vectors for the largest Fisher gain over the full-sample 2PCF

ASTRA clustering project

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Abstract

Using the complete grid of full-box runs on disk (nine AbacusSummit cosmologies $\times$ 50 HOD draws), we ask which ASTRA environment-split statistic, or which pair of statistics, extracts the most cosmological Fisher information beyond the plain full-sample auto-correlation. We sweep all 16 single quantile vectors and all 120 pairs, in two framings (standalone and added to the full auto), scoring each by the HOD-marginalised Fisher figure of merit. The naive ranking is dominated by the *random*-quantile legs, with apparent gains up to $55\times$; we show this is an artefact of unmodelled derivative noise in those noise-limited legs (all of the global top 20 carry a random leg), not real information. Among the trustworthy *data*-leg vectors the gains are real: adding a single full $\times$ data-quantile cross buys $\sim 5\times$ in the three-parameter FoM, and a greedy forward selection that spans the four density quantiles reaches a five-stem vector (`full+xdQ1+dQ1+dQ2+xdQ3+dQ4`) with $\text{FoM}_3 = 43\times$ — marginalised errors tighter than the standard 2PCF by $3.4\times$ (ω_b), $4.1\times$ (ω_c), $3.6\times$ (n_s) and $2.8\times$ (σ_8), with clear diminishing returns past four stems. We document the method, the artefact diagnosis, the honest ranking, and the greedy chain.

1 Question and data

ASTRA splits a galaxy sample into four environment quantiles (Q1 underdense $\rightarrow$ Q4 overdense) and measures, per box, the multipoles $\xi_{0,2}(s)$ of: the full-sample auto-correlation (`tpcf_full_data`, our baseline), the four data-quantile autos (`dQ1..4`), the four random-quantile autos (`rQ1..4`), the four full $\times$ data-quantile crosses (`xdQ1..4`), and the four full $\times$ random-quantile crosses (`xrQ1..4`). That is $17 \text{ stems} \times \{\ell = 0, 2\} \times 15 \text{ bins per run}$.

The signal data are the $9 \times 50 = 450$ phase-matched full-box runs (`data/fullbox/`); cosmology derivatives come from the global linear response fit (`derivative_global_*`, condition number 1.34 on the full grid). The covariance of one $500 h^{-1} \text{ Mpc}$ subbox is estimated from the mean-subtracted subboxes pooled across all nine cosmologies, $9 \times 64 = 576$ fluctuation samples, and scaled to the full $2000 h^{-1} \text{ Mpc}$ box volume ($C/64$). The HOD nuisance covariance C_{HOD} is the empirical scatter of the 50 same-phase `c000` draws.

2 Method

The analysis is implemented in `scripts/fisher_vector_search.py`, which reuses the assembly and Fisher machinery of `scripts/fisher_joint.py`. The steps are:

1. **Candidate vectors.** Each candidate is one or two ASTRA stems, taken at *native* 15-bin resolution in mono+quad ($\ell = 0, 2$). The 576-sample pooled covariance keeps the Hartlap

Table 1: Raw top of the additive ranking (full auto + stem(s)). A star marks a candidate that contains a random leg (rQ or xrQ). *Every* entry in the global top eight, and all 20 of the global top 20, carry a random leg, despite random legs being the weakest tracers in every earlier single-parameter test.

	vector	FoM ₃ gain	FoM ₄ gain
singles	xrQ4 [*]	12.2	13.5
	rQ4 [*]	8.6	9.9
	xrQ1 [*]	7.8	8.6
pairs	xrQ1+xrQ4 [*]	55.8	64.6
	xdQ1+xrQ4 [*]	44.6	59.0
	xdQ4+xrQ4 [*]	43.4	60.3

factor benign even at $n_b = 60$ (pair) or 90 (pair added to the full auto), so no rebinning is needed. We evaluate the 16 singles and all $\binom{16}{2} = 120$ pairs.

2. **Two framings.** *Standalone* scores the candidate by itself against the full auto (“is it better than the 2PCF?”). *Additive* scores the full auto with the candidate appended (“what does ASTRA add to the 2PCF?”). The additive framing is the physical measure of marginal information and is our headline.
3. **Fisher and marginalisation.** For each candidate we build the 16×16 joint Fisher over the four cosmological parameters $\{\omega_b, \omega_c, n_s, \sigma_8\}$ and the twelve yuan23 HOD parameters, $F = D C^{-1} D^T$ with Hartlap-corrected precision, add a Gaussian yuan23 prior on the HOD block, and read off the HOD-marginalised 4×4 cosmology covariance $\text{Cov} = (F + F_{\text{prior}})_{[\text{cosmo}]}^{-1}$ (route a).
4. **Figure of merit.** We reduce Cov to

$$\text{FoM}_4 = [\det \text{Cov}_4]^{-1/2}, \quad \text{FoM}_3 = [\det \text{Cov}_3]^{-1/2}, \quad (1)$$

where Cov₃ drops σ_8 (the upper-left 3×3 block, already marginalised over σ_8). The *gain* of a candidate is $\text{FoM}/\text{FoM}_{\text{full auto}}$. We headline FoM₃ because the σ_8 derivative is unreliable (Sec. 3), so a four-parameter FoM inherits that unreliability.

3 The random-leg trap

Table 1 and the bright rQ4/xrQ4/xrQ1 band in Fig. 2 show that the naive sweep is dominated by random legs, with implausible gains (FoM₃ up to 55 $\times$, i.e. a factor ~ 4 on every parameter from one box). This contradicts every prior result, in which the underdense random leg was the *weakest* vector and carried the most derivative noise. The resolution is that these gains are an artefact:

- The random quantiles — especially their quadrupoles — are *noise-limited*: their per-bin ASTRA-random scatter is comparable to the HOD-induced signal (measured separately: rQ2/3 $\ell=2$ noise/ signal ≈ 0.7 – 0.8). Their global-fit derivatives therefore contain large noise components.

Table 2: Data-leg candidates only (no random leg). Additive framing, HOD-marginalised. Baseline full-auto errors: $\sigma(\omega_b) = 5.7 \times 10^{-4}$, $\sigma(\omega_c) = 1.7 \times 10^{-3}$, $\sigma(n_s) = 1.3 \times 10^{-2}$, $\sigma(\sigma_8) = 1.4 \times 10^{-2}$.

	vector	FoM ₃ gain	$\sigma(\omega_b)$	$\sigma(\omega_c)$	$\sigma(n_s)$
singles	xdQ1	5.1	3.3×10^{-4}	8.4×10^{-4}	8.3×10^{-3}
	xdQ4	5.0	3.2×10^{-4}	8.4×10^{-4}	8.5×10^{-3}
	dQ4	4.1	3.3×10^{-4}	9.8×10^{-4}	8.4×10^{-3}
	dQ1	3.9	4.2×10^{-4}	1.1×10^{-3}	7.7×10^{-3}
pairs	dQ1+xdQ1	14.0	2.8×10^{-4}	—	6.0×10^{-3}
	dQ2+xdQ1	12.8	2.5×10^{-4}	—	4.9×10^{-3}
	xdQ3+xdQ4	12.2	2.6×10^{-4}	—	5.0×10^{-3}

- A Fisher matrix treats the derivative D as exact. Noise in D that is not represented in the data covariance C inflates $D C^{-1} D^T$: the Fisher reads derivative noise as constraining power, and a noisy derivative pointing in a direction nearly orthogonal to the full auto spuriously “breaks” degeneracies.
- The signature is diagnostic: a random leg is barely better than the full auto on its own (**xrQ4** standalone FoM₃ = 1.6) yet explodes when appended (12×), and *all* 20 top-20 vectors carry a random leg. Real information would not concentrate so completely in the lowest-S/N legs.

We therefore flag every random-leg candidate and base the physical conclusions on the data legs only. Removing the random-leg artefact is exactly the role of propagating the emulator derivative variance (future work); it would deflate these spurious gains.

4 The honest, data-leg ranking

Restricting to data-leg candidates (Table 2, Fig. 1) gives a clean and physically sensible picture:

- **Singles.** The full×data-quantile crosses at the extremes lead — **xdQ1** (void) and **xdQ4** (knot), FoM₃ gain ≈ 5 — followed by the extreme data autos **dQ4**, **dQ1** (≈ 4). The middle quantiles (Q2, Q3) are weaker. Each of these improves ω_b and n_s by ~ 1.6 – $1.7\times$ over the full auto. Note that on its own each vector is *worse* than the full auto (standalone FoM₃ < 1.1); the gain is entirely degeneracy-breaking when added — the ASTRA thesis.
- **Pairs.** The best clean pair is **dQ1+xdQ1** (void data auto + void data cross), FoM₃ gain 14, i.e. a factor ~ 2.0 on ω_b and ~ 2.2 on n_s over the full auto. Combinations of an extreme data auto with an extreme data cross dominate; pairs of two middle quantiles, or two stems from the same family, add little (redundant).
- **σ_8 is untouched.** No vector improves σ_8 appreciably ($1.4 \times 10^{-2} \rightarrow \sim 1.0 \times 10^{-2}$), consistent with σ_8 being amplitude-degenerate with the HOD and with its derivative being unreliable. This is why FoM₃ (which drops σ_8) is the trustworthy score; Fig. 3 confirms the per-parameter picture.

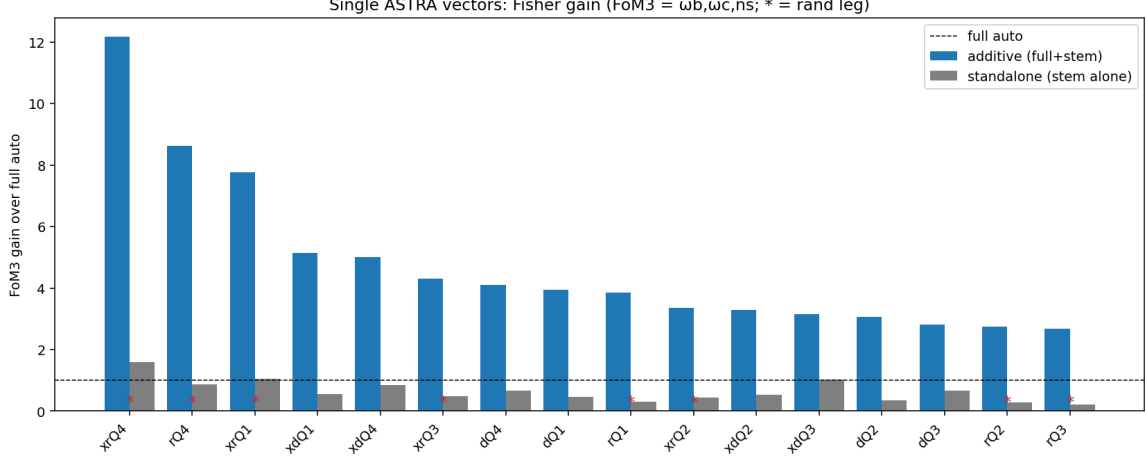


Figure 1: Single-vector FoM₃ gain over the full auto, additive (blue) and standalone (grey), ranked. Red stars mark random-leg vectors, whose additive gains are inflated by derivative noise (Sec. 3). Every vector is at or below the full-auto line on its own; the gain is degeneracy-breaking, realised only when appended.

Table 3: Greedy chain over the data-leg stems. “marg.” is the multiplicative gain in FoM₃ from that one step. σ values are the HOD-marginalised errors of the cumulative vector. The chain visits all four density quantiles by step five.

step	add	meaning	FoM ₃	marg.	$\sigma(\omega_b)$	$\sigma(n_s)$	$\sigma(\sigma_8)$
0	—	full auto	1.0	—	5.7×10^{-4}	1.3×10^{-2}	1.4×10^{-2}
1	xdQ1	void cross	5.1	5.1	3.3×10^{-4}	8.3×10^{-3}	1.0×10^{-2}
2	dQ1	void auto	14.0	2.7	2.8×10^{-4}	6.0×10^{-3}	7.5×10^{-3}
3	dQ2	low–mid auto	24.7	1.8	2.3×10^{-4}	4.3×10^{-3}	6.8×10^{-3}
4	xdQ3	mid–high cross	32.5	1.3	2.1×10^{-4}	3.8×10^{-3}	5.4×10^{-3}
5	dQ4	knot auto	43.3	1.3	1.7×10^{-4}	3.6×10^{-3}	5.0×10^{-3}

5 Greedy forward selection: how many stems, and which?

The single- and pair-level results raise the obvious question: if one extreme data cross already buys $\sim 5\times$ and the void auto+cross pair buys $14\times$, how far does it go? We answer it with a *greedy forward selection* restricted to the eight data-leg stems (`scripts/fisher_greedy_chain.py`). Starting from the full auto, at each step we append the single data-leg stem that most increases FoM₃, keep it, and repeat, up to five added stems. Greedy selection is not guaranteed to find the globally optimal subset, but it is the natural way to read off the *marginal* value of each successive statistic and to expose redundancy.

The chain is `full+xdQ1+dQ1+dQ2+xdQ3+dQ4` (Table 3, Fig. 4). Three things stand out.

- **Strong diminishing returns.** The per-step gain falls $5.1 \rightarrow 2.7 \rightarrow 1.8 \rightarrow 1.3 \rightarrow 1.3$. The first two stems carry most of the information (FoM₃ reaches 14 of the eventual 43 by step two); beyond step four each new statistic adds only $\sim 30\%$, while the bin count and Hartlap/redundancy cost keep rising. A four-stem vector is the practical sweet spot.
- **Gains come from spanning environments, not repetition.** Greedy does not pile up void

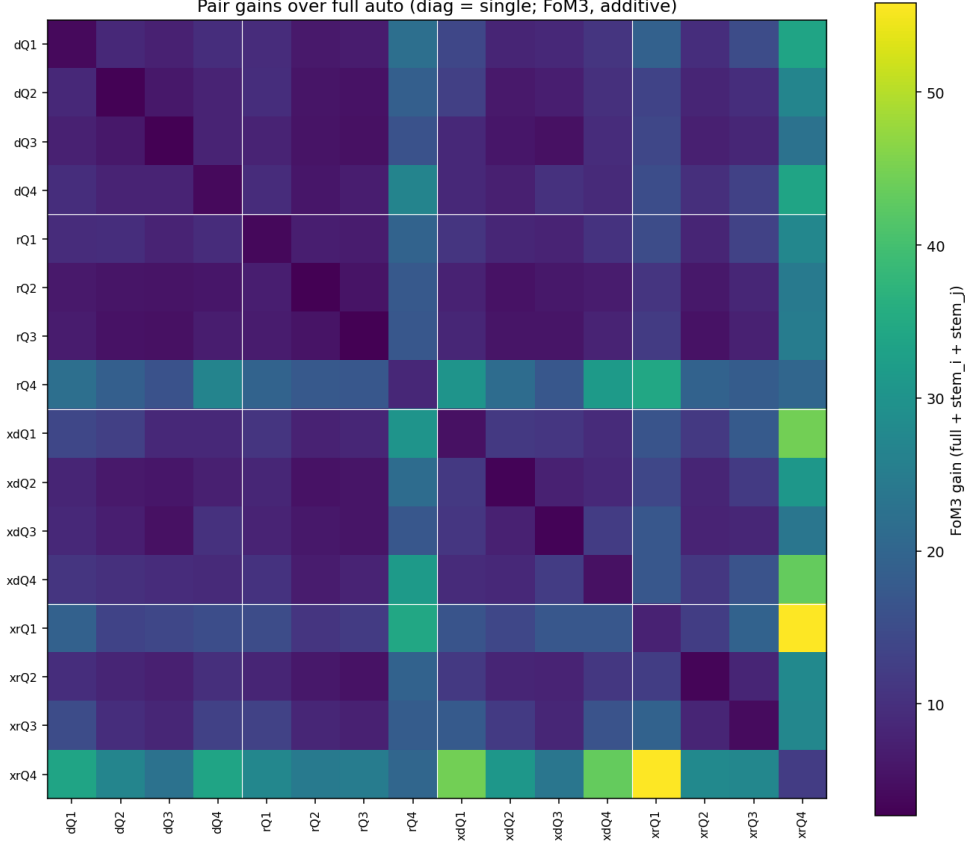


Figure 2: Additive FoM₃ gain for every pair, full auto + stem_i + stem_j (diagonal = single). White lines separate the four families (dQ, rQ, xdQ, xrQ). The bright band on the rQ4/xrQ4/xrQ1 rows and columns is the random-leg artefact; the data-leg quadrants (dQ, xdQ) show the modest, trustworthy gains.

statistics; after the void auto+cross it reaches for a *different* density bin (dQ2), then mid-high (xdQ3), then the knot (dQ4), touching all four quantiles by step five. Different environments respond to cosmology and to the HOD in different ways, so each new environment breaks a *complementary* degeneracy; a second statistic from the same environment would be largely redundant.

- **σ_8 moves only in combination.** No single vector improved σ_8 (Sec. 4), but the cumulative five-stem vector tightens it $2.8 \times (1.4 \times 10^{-2} \rightarrow 5.0 \times 10^{-3})$. The multi-environment combination partially lifts the amplitude degeneracy that defeats any one statistic — though σ_8 remains the least trustworthy entry because its derivative is unreliable, and FoM₃ (which excludes it) stays our headline score.

The five-stem vector versus the 2PCF

Figure 5 is the head-to-head Fisher comparison of the standard full-sample 2PCF (tpcf_full_data, mono+quad) and the five-stem ASTRA vector, both HOD-marginalised, generated by scripts/fisher.compare.fu. The five-stem 68/95% contours sit nested well inside the full-auto contours on every panel, with the marginalised errors tighter by

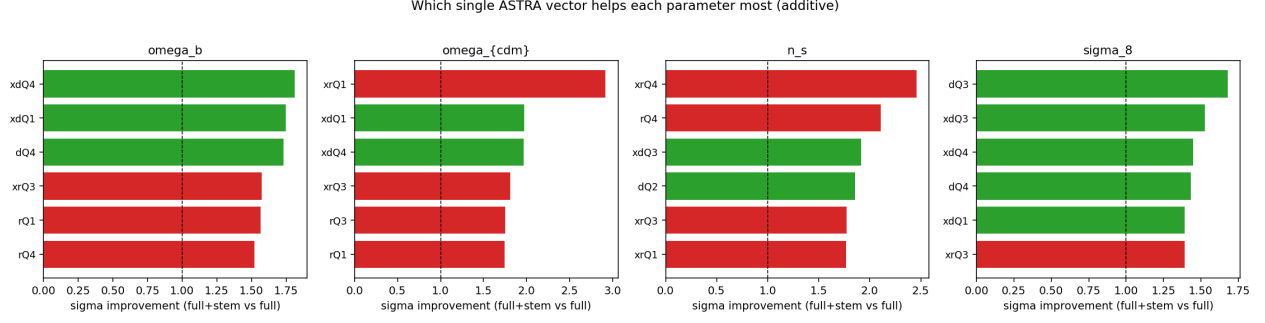


Figure 3: For each cosmological parameter, the single ASTRA vectors that most improve its marginalised error when added to the full auto. Green bars are data-leg vectors (trustworthy); red bars are random-leg vectors (artefact).

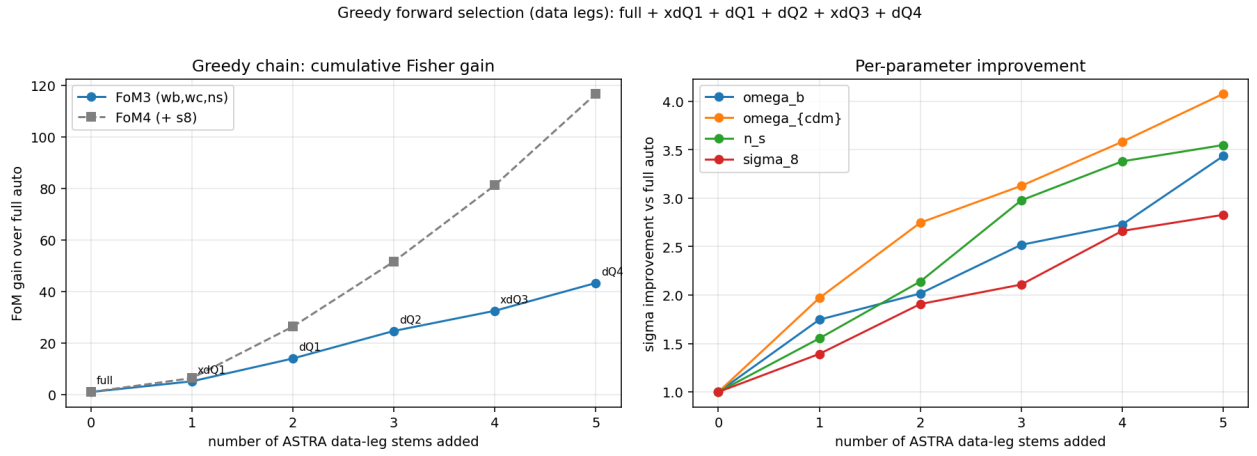


Figure 4: Greedy forward selection over the data legs. Left: cumulative FoM_3 (and FoM_4) gain over the full auto versus the number of ASTRA stems added — the curve flattens after ~ 4 stems. Right: per-parameter σ improvement over the full auto. Labels mark the stem chosen at each step.

$$\sigma(\omega_b) \ 3.4\times, \quad \sigma(\omega_c) \ 4.1\times, \quad \sigma(n_s) \ 3.6\times, \quad \sigma(\sigma_8) \ 2.8\times, \quad (2)$$

i.e. $\text{FoM}_3 = 43\times$. Crucially the blue ellipses are not merely smaller but *rotated* relative to grey: the environment splits change the orientation of the parameter degeneracies, which is the geometric signature of genuinely new information rather than a uniform rescaling. The five-stem 95% region is inside the full-auto 68% region for most parameter pairs — a qualitatively different constraint from a single $2h^{-1}$ Mpc box, by reanalysis alone.

6 Anatomy of the 5-stem Fisher computation

This section opens up the Fisher calculation behind Fig. 5 step by step, so every ingredient is visible. The five-stem vector concatenates six pieces (the full auto plus the five selected stems), each in mono+quad at the native 15-bin resolution, for a data vector of $n_b = 6 \times 2 \times 15 = 180$ bins. All figures are produced by `scripts/fisher_5stem_details.py`.

The Fisher matrix is

$$F = D C^{-1} D^T, \quad (3)$$

where D is the $(4 + 12) \times n_b$ matrix of derivatives of the data vector with respect to the four cosmological and twelve HOD parameters, and C is the data covariance. The cosmology constraints follow from inverting either the cosmology block alone (conditional, HOD fixed) or the full matrix plus the HOD prior and taking the cosmology sub-block (marginalised). The four steps below are the four factors in Eq. (3).

Step 1 — the data vector

Figure 6 shows the fiducial (c000 ensemble-mean) data vector, one panel per piece. The monopole amplitude grows monotonically from the void (**dQ1**) to the knot (**dQ4**, an order of magnitude larger): the overdense quantile is the most strongly clustered and highest-bias tracer, the underdense one the least. The quadrupoles ($\ell = 2$) carry the redshift-space anisotropy. These six curves, stacked $[\ell 0, \ell 2]$ per piece, are the 180-component vector whose information content we are dissecting.

Step 2 — the derivatives D

Figure 7 shows the four cosmology rows of D (the global-response derivatives $\partial\xi/\partial\theta$) across all 180 bins, with the six piece blocks marked. Two features matter for the Fisher information:

- The derivative amplitude is largest on the high-bias **dQ4** piece, especially at small s — the knot quantile is the most cosmology-responsive.
- The *shape* of the response differs from piece to piece (e.g. the ω_b acoustic feature versus the broadband n_s tilt appear with different relative weights in each environment). It is precisely this piece-to-piece difference in $\partial\xi/\partial\theta$ that lets the environment splits break parameter degeneracies the full auto cannot.

(The twelve HOD rows of D , not shown, enter the marginalisation in Step 4.)

Step 3 — the covariance C

Figure 8 is the 180×180 correlation matrix of the data vector, estimated from the mean-subtracted subboxes pooled across all nine cosmologies (576 samples) and scaled to the full-box volume. The block structure is the crux of how much *independent* information the vector holds:

- Within each piece, neighbouring s -bins are positively correlated, and the monopole and quadrupole blocks are cross-correlated.
- Across pieces, the full auto is strongly correlated with the full×data crosses (**xdQ1**, **xdQ3**) because they literally share the full-sample leg, and the quantile autos are mutually correlated through the shared density field. These off-diagonal correlations are why the gains saturate: the pieces are not independent, and C^{-1} correctly down-weights the redundancy.

The 576 pooled samples are what make $n_b = 180$ affordable: the Hartlap factor is $(576 - 180 - 2)/(576 - 1) = 0.685$, comfortably positive, where the 64-subbox estimate would have failed ($n_b > 64$).

Step 4 — the Fisher matrix and parameter covariance

Combining D and C via Eq. (3) with the Hartlap-corrected precision gives the parameter covariance. Figure 9 shows the 4×4 cosmology parameter correlation matrices, conditional (HOD fixed) and marginalised (HOD nuisances marginalised with the yuan23 prior), with the per-parameter σ in the titles:

	$\sigma(\omega_b)$	$\sigma(\omega_c)$	$\sigma(n_s)$	$\sigma(\sigma_8)$
conditional (HOD fixed)	2.9×10^{-5}	2.0×10^{-4}	7.6×10^{-4}	2.9×10^{-3}
marginalised	1.7×10^{-4}	4.1×10^{-4}	3.6×10^{-3}	5.0×10^{-3}

Two lessons:

- **Marginalising over the HOD costs a factor few** per parameter ($6\times$ on ω_b , $5\times$ on n_s): the cosmology directions that overlap the HOD response are degraded most. This is the honest price of not knowing the galaxy–halo connection, and it is the regime all our headline numbers are quoted in.
- **Marginalisation re-shapes the degeneracies.** Conditionally, ω_b is strongly anti-correlated with σ_8 (-0.84) and ω_c with n_s (-0.79); after HOD marginalisation these soften and new correlations appear ($\omega_b\text{--}\omega_c = +0.55$, $n_s\text{--}\sigma_8 = -0.42$). The rotation of the ellipses in Fig. 5 is exactly this change of correlation structure.

7 Conclusions

1. The systematic sweep of all 16 singles and 120 pairs is dominated, at face value, by random-leg vectors with gains up to $55\times$. This is a **derivative-noise artefact**, not physics: the noise-limited random legs have noisy global-fit derivatives that the Fisher misreads as information. Any data-vector ranking on this grid must exclude or noise-correct the random legs.
2. Among the **trustworthy data legs**, the full $\times$ data-quantile crosses at the density extremes (xdQ1, xdQ4) are the best single additions (FoM₃ gain ≈ 5), and the best clean pair is the void data auto with the void data cross, dQ1+xdQ1 (FoM₃ gain 14, $\sim 2.4\times$ per parameter on ω_b, ω_c, n_s). The gains are pure degeneracy breaking: each vector is worse than the full auto alone.
3. A **greedy chain** over the data legs (full+xdQ1+dQ1+dQ2+xdQ3+dQ4) reaches FoM₃ = $43\times$ at five stems, tightening the standard-2PCF errors by $3.4\text{--}4.1\times$ on ω_b, ω_c, n_s and $2.8\times$ on σ_8 (Fig. 5). Returns diminish sharply ($5.1 \rightarrow 2.7 \rightarrow 1.8 \rightarrow 1.3$ per step), so ~ 4 stems is the practical optimum, and the gains come from *spanning* the four environments rather than repeating one.
4. Single vectors do not improve σ_8 (HOD-amplitude degeneracy, unreliable derivative), but the multi-environment combination does so modestly; FoM₃ excluding σ_8 remains the trustworthy score.
5. **Next steps.** (i) Propagate the GP emulator derivative variance (calibrated with the $\sim 1.25\times$ inflation measured separately) so the random-leg artefact is deflated and the random legs can be ranked honestly; (ii) use the higher-iteration runs to denoise the random-quantile derivatives directly; (iii) revisit the four-parameter FoM once a trustworthy σ_8 derivative is available.

All numbers are reproduced by `scripts/fisher_vector_search.py` (the full sweep; ranked table in `plots/vector_search/vector_search_results.csv`), `scripts/fisher_greedy_chain.py` (the greedy chain), and `scripts/fisher_compare_full_vs_5stem.py` (the corner comparison), and `scripts/fisher_5stem_details.py` (the step-by-step anatomy of Sec. 6).

Fisher 68/95%: full-auto 2PCF vs 5-stem ASTRA (full + xdQ1 + dQ1 + dQ2 + xdQ3 + dQ4), HOD-marginalised

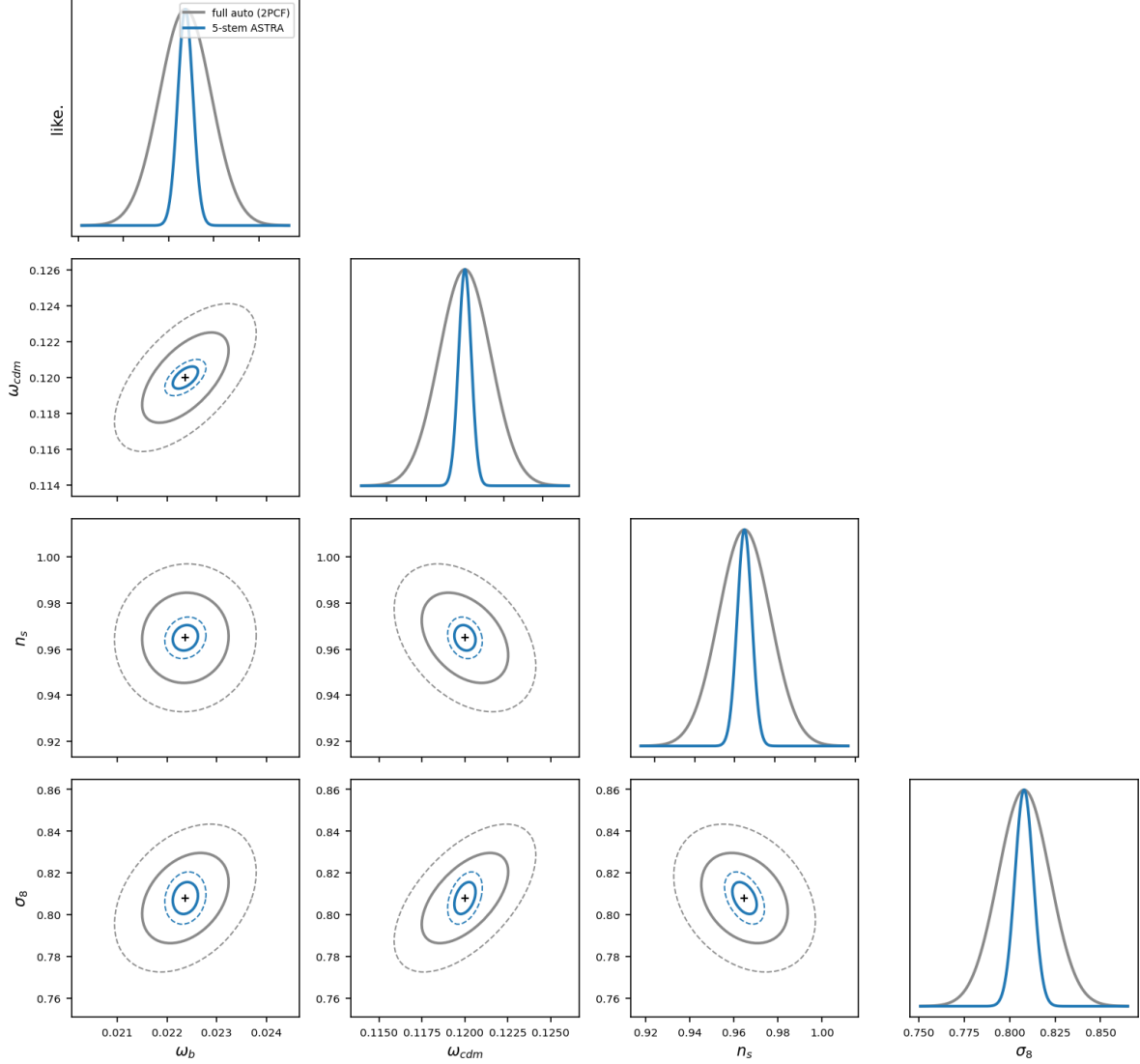


Figure 5: HOD-marginalised Fisher constraints: full-auto 2PCF (grey) versus the five-stem ASTRA vector **full+xdQ1+dQ1+dQ2+xdQ3+dQ4** (blue), 68% (solid) and 95% (dashed). The blue contours are both smaller and rotated — ASTRA shrinks and re-orientes the cosmology degeneracies.

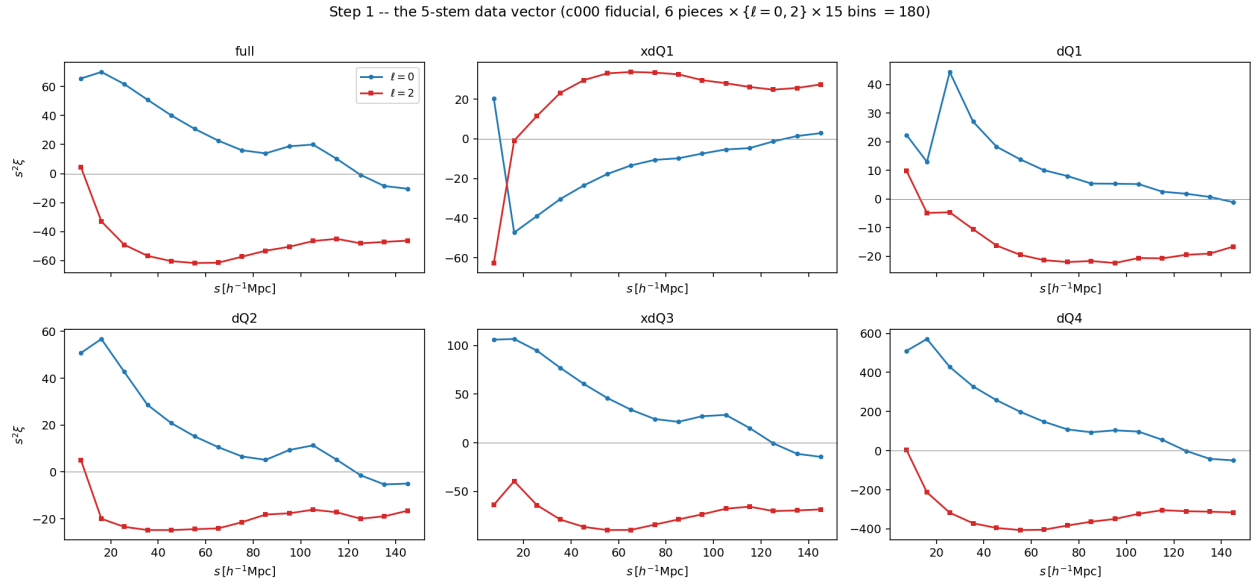


Figure 6: Step 1: the fiducial five-stem data vector, $s^2 \xi_{0,2}(s)$ per piece. Amplitude grows void (dQ1) $\rightarrow$ knot (dQ4).

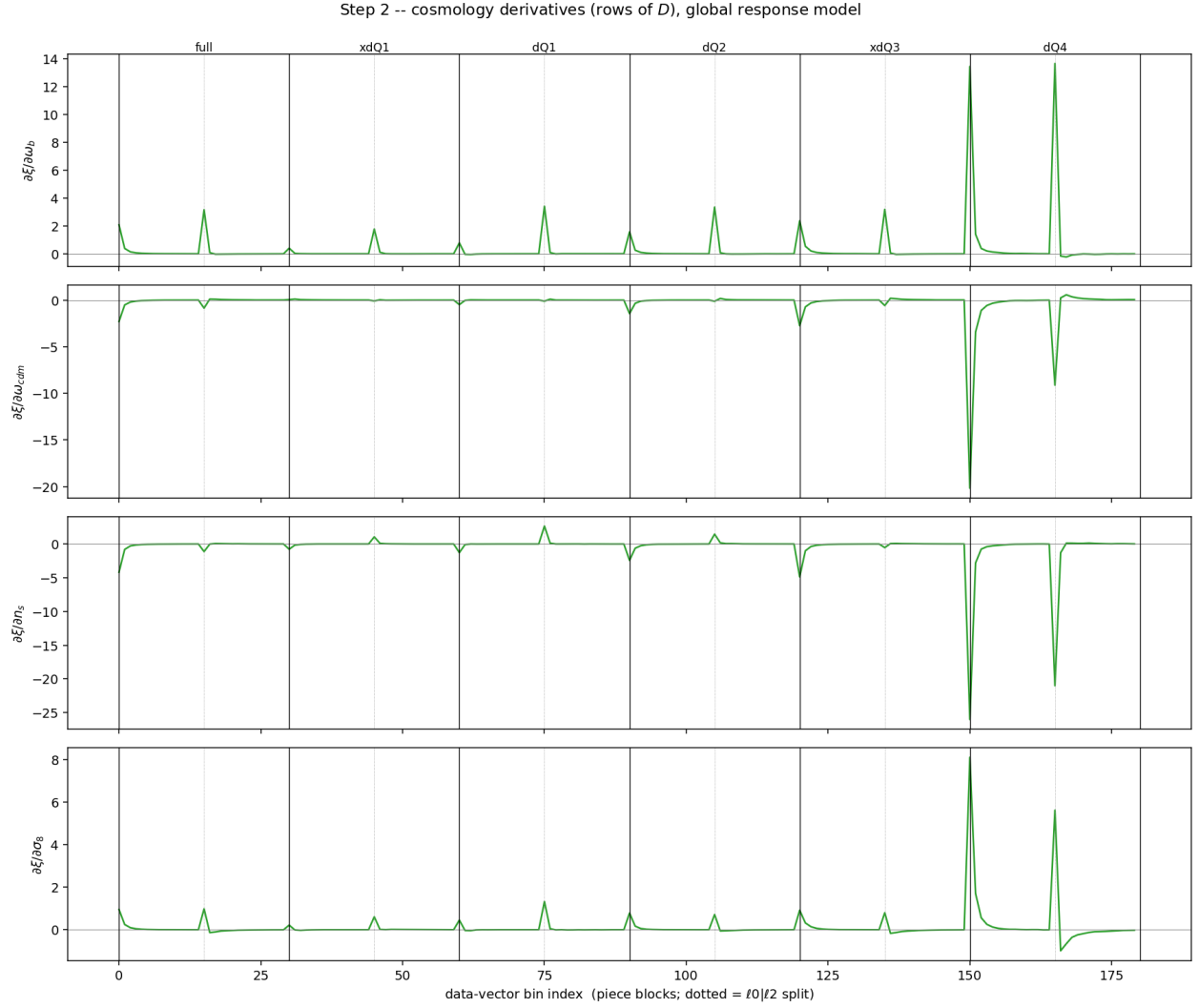


Figure 7: Step 2: the four cosmology rows of D , $\partial\xi/\partial\theta$, across the 180-bin vector. Solid lines separate the six pieces; dotted lines mark the $\ell_0|\ell_2$ split within each piece. The high-bias **dQ4** piece carries the largest response.

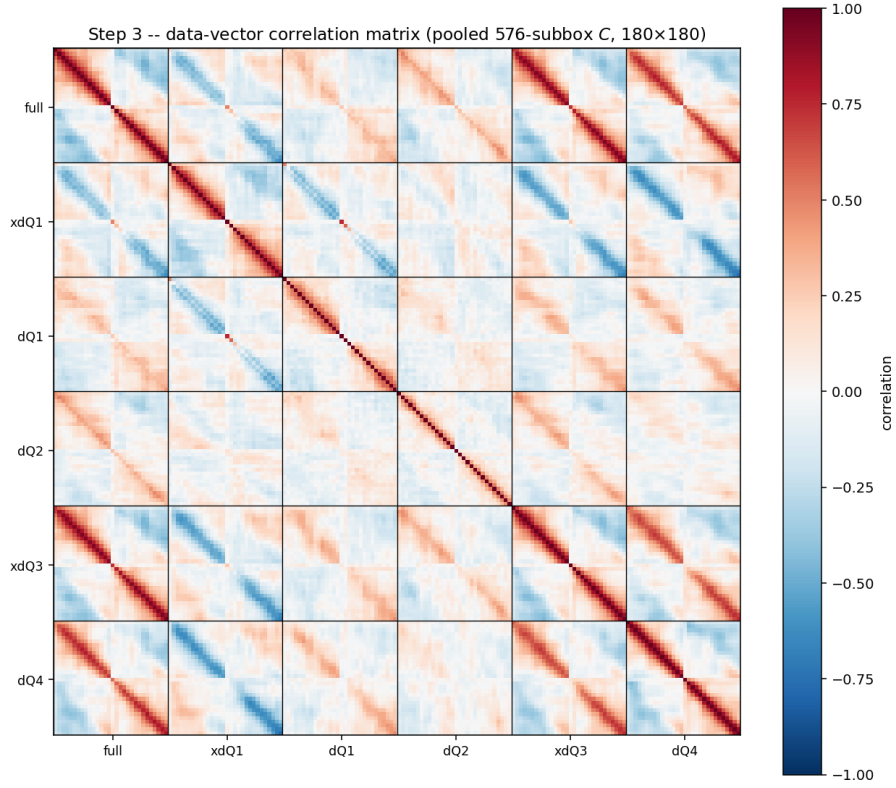


Figure 8: Step 3: the 180×180 data-vector correlation matrix (pooled 576-subbox C). Black lines separate the six pieces. Strong cross-piece correlations (e.g. full with the xd crosses) are the redundancy that limits the cumulative gain.

Step 4 -- cosmology parameter correlations from $F = DC^{-1}D^T$ (Hartlap 0.69)

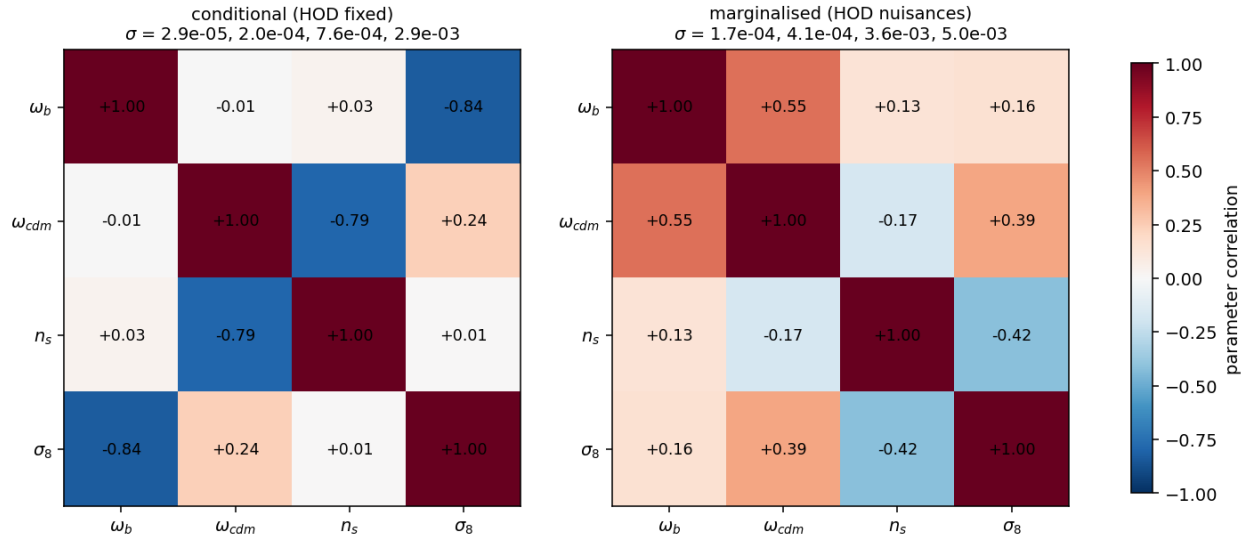


Figure 9: Step 4: cosmology parameter correlation matrices from $F = DC^{-1}D^T$, conditional (left) and HOD-marginalised (right), with the per-parameter σ in the titles. HOD marginalisation both inflates the errors and re-shapes the degeneracies.